BIOCHEMISTRY CUMULATIVE EXAMINATION

April 6, 1991

(50) If the following DNA primer and template molecules are used in a standard set of four DNA polymerase-catalyzed "dideoxy-" chain-termination sequencing reactions, with the label introduced as α-^{32}P-dCTP, give the expected pattern of bands to be observed in an autoradiogram following gel electrophoresis and autoradiography:

5'-GGCCTTATGG-3'

5'-TCGGACAAGCGCACATGTGGAATGACCATAAGGCC-3'

(25) If 1 pmole of the above template and 400 pmoles of the above primer are used for 4 cycles of the polymerase chain reaction (PCR), list the sequences and amounts of the DNA molecules expected to be present in the sample after 4 cycles, assuming ideal 100% yields for each cycle.

(25) If 400 pmoles of a second primer, 5'-TCGAACAAGC-3', are used together with 1 pmole of the above template and 400 pmoles of the above primer for 4 cycles of PCR, list the sequences and amounts of all of the DNA molecules expected to be present in the sample after 4 cycles, assuming ideal 100% yields.